

NANOPARTICLE-MEDIATED MEMBRANE FUSION: INTERPLAY OF MEMBRANE CURVATURE AND NANOPARTICLE SIZE

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Keywords: membrane fusion, amphiphilic nanoparticle, lipid vesicle

ABSTRACT

The ability to induce membrane fusion in vitro has become a major goal in synthetic biology. Amphiphilic gold nanoparticles (AuNPs) embedded in zwitterionic lipid vesicles can act as a minimal synthetic fusion machinery tuned by calcium ions. However, very little is known about the physical determinants that regulate this fusion process. Here we investigate the roles of lipid membrane curvature and amphiphilic nanoparticle size. We used AuNPs with the same amphiphilic coating but different core diameters (~2 nm and ~4 nm), interacting with DOPC/cholesterol unilamellar vesicles of different membrane curvatures. Using fluorescence spectroscopy assays, dissipative quartz microbalance, and molecular dynamics simulations, we found that small AuNPs promote vesicle fusion regardless of the membrane curvature, while large AuNPs, hindering the progression from the stalk state to the hemifusion state, are not fusogenic with low curvature membranes and can induce fusion events only with significantly curved membranes. These findings offer novel insights into the rational design of synthetic fusion strategies.

REFERENCES

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